

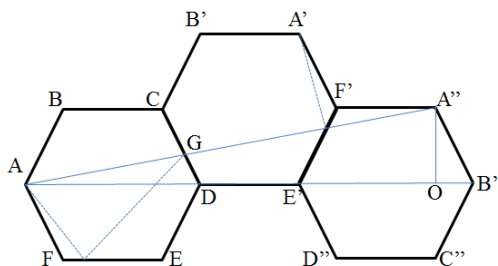
ARML Local 2017 Team Solutions

T-1. The numbers between 1 and 23, inclusive, can be the sum of at most 3 non-zero die rolls (1, 1 + 1, 3, 3 + 1, 3 + 1 + 1, 6, 6 + 1, 6 + 1 + 1, 6 + 3, 10, 10 + 1 + 1, 6 + 6, 10 + 3, 10 + 3 + 1, 6 + 6 + 3, 10 + 6, 10 + 6 + 1, 6 + 6 + 6, 10 + 6 + 3, 10 + 10, 10 + 10 + 1, 10 + 6 + 6, 10 + 10 + 3). The numbers 1 through 33 inclusive can be achieved via four rolls by adding an additional roll of 10 to the up to three rolls that sum to 14 to 23. Accordingly, $\boxed{34}$ is the smallest positive integer that cannot be the sum of four rolls of this die.

T-2. If each factorial is expanded out to its individual terms, there would be 96 5s, 91 10s, ..., 6 95s, and 1 100 terms. Each multiple of five term adds one zero to the end of z_{100} , with each multiple of 25 contributing an additional zero to the end of z_{100} . The total number of zeros is $(96 + 91 + \cdots + 6 + 1) + (76 + 51 + 26 + 1) = \boxed{1124}$.

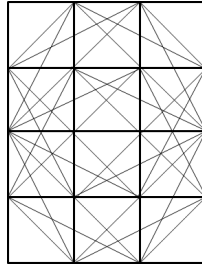
T-3. Put $A = (0, 0)$, let $B = (5, y)$ and $C = (x, 5)$. Then the area of ABC is equal to $\frac{1}{2} \begin{vmatrix} 1 & 0 & 0 \\ 1 & 5 & y \\ 1 & x & 5 \end{vmatrix} = \frac{1}{2}(25 - xy)$. Because x and y are independent, $E[xy] = E[x]E[y] = (5/2)^2 = 25/4$. The expected area of ABC is $\frac{1}{2} \left(25 - \frac{25}{4} \right) = \boxed{\frac{75}{8}}$.

T-4. Reflect the hexagon $ABCDEF$ across the segment $\overline{CD}$ to get the hexagon $A'B'CDE'F'$. In order to strike exactly one additional side and return to A , the laser should strike $\overline{EF}$ next before returning to A . Reflecting the second hexagon about $\overline{E'F'}$ to get the hexagon $A''B''C''D''E'F'$, the segment $\overline{AA''}$ traces the path of the laser beam when reflected back upon the initial hexagon as shown below.



Dropping the perpendicular from A'' to $\overline{AB''}$ at O , $AO = \frac{9}{2}AB$ and $A''O = \frac{\sqrt{3}}{2}AB$, and $\tan(\angle DAG) = \tan(\angle OAA'') = \frac{\sqrt{3}/2}{9/2} = \boxed{\frac{\sqrt{3}}{9}}$.

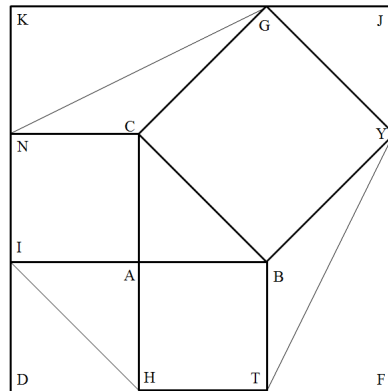
- T-5. In the 3×4 grid, there are 12 squares with side length 1, 6 of side length 2, and 2 of side length 3. There are also squares with edges not aligned with the grid; 6 of side length $\sqrt{2}$ and 4 of side length $\sqrt{3}$, for a total of $\boxed{30}$.



- T-6. Multiplying both sides of the equation by $\frac{\sqrt{2}}{2}$, $\frac{\sqrt{2}}{2} \sin 15^\circ + \frac{\sqrt{2}}{2} \cos 15^\circ = \sin A \rightarrow \sin A = \cos 45^\circ \sin 15^\circ + \sin 45^\circ \cos 15^\circ = \sin(45^\circ + 15^\circ) = \sin(60^\circ)$. Because A is acute, $A = 60^\circ$, so $\cos A = \boxed{\frac{1}{2}}$.

- T-7. For this process, all integers up to 24 return 1. The integers 25 to 29 return 2. The smallest integer for which the product of its digits is between 25 and 29, inclusive, is $\boxed{39}$.

- T-8. As the three squares have side lengths of 4, 4, and $4\sqrt{2}$, it is clear that ABC is an isosceles right triangle. Let $DFJK$ be the square of side length 12 for which $\overline{HT}$ is a subsegment of $\overline{DF}$ and $\overline{IN}$ is a subsegment of $\overline{DK}$. Then $[THINGY] = [DFJK] - [HID] - [FTY] - [JGY] - [GNK] = 144 - 8 - 16 - 8 - 16 = \boxed{96}$.



- T-9. There are five ways to represent x^8 as the product of exactly eight terms of the form x^2 , x , and 1: $(x^2)^4(x)^0(1)^4$, $(x^2)^3(x)^2(1)^3$, $(x^2)^2(x)^4(1)^2$, $(x^2)^1(x)^6(1)^1$, and $(x^2)^0(x)^8(1)^0$. The coefficient of x^8 will be $\binom{8}{4,0,4} + \binom{8}{3,2,3} + \binom{8}{2,4,2} + \binom{8}{1,6,1} + \binom{8}{0,8,0} = \frac{8!}{4!0!4!} + \frac{8!}{3!2!3!} + \frac{8!}{2!4!2!} + \frac{8!}{1!6!1!} + \frac{8!}{0!8!0!} = 70 + 560 + 420 + 56 + 1 = \boxed{1107}$.

- T-10. First place 1 and 8 in the upper-left and bottom-right. WLOG 2 is the entry to the right of 1 (by symmetry the final answer is twice the number of configurations with 2 to the right of 1). Then, the grid is filled in as follows:

1	2	a
x		b
y	z	8

Observe that $x \leq y \leq z$, $a \leq b$, and $x \leq b$. There are $\binom{5}{2}$ ways to pick a and b , and all of them result in valid configurations except $(a, b) = (3, 4)$. Hence the final answer is $2\left(\binom{5}{2} - 1\right) = \boxed{18}$.

- T-11. Rolling the dice in sequence, it is clear to see that the probability that all four die rolls are different is $\frac{6 \cdot 5 \cdot 4 \cdot 3}{6 \cdot 6 \cdot 6 \cdot 6} = \frac{5}{18}$. The game is fair provided $\frac{5}{18}(1) + \frac{13}{18}(-k) = 0 \rightarrow k = \boxed{\frac{5}{13}}$.

- T-12. Note that while b must be positive, a may be negative. The two equations in the problem are equivalent to $2\log_2(a^2) - 3\log_2(b) = 3$ and $2\log_2(a^2) + 3\log_2(b) = 9$. Adding the two equations together gives $4\log_2(a^2) = 12 \rightarrow \log_2(a^2) = 3 \rightarrow \log_2(b) = 1$. Thus $b = 2$ and $a = \pm 2\sqrt{2}$, so the minimum value of ab is $\boxed{-4\sqrt{2}}$.

- T-13. $\frac{20}{(n-2)(n+2)} = \frac{5}{n-2} - \frac{5}{n+2}$, so

$$\begin{aligned}
 \sum_{n=3}^{10} \frac{20}{(n-2)(n+2)} &= 5 \left(\sum_{n=3}^{10} \frac{1}{n-2} - \sum_{n=3}^{10} \frac{1}{n+2} \right) \\
 &= 5 \left(\left(\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{8} \right) - \left(\frac{1}{5} + \frac{1}{6} + \cdots + \frac{1}{12} \right) \right) \\
 &= 5 \left(\left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) - \left(\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} \right) \right) \\
 &= 5 \left(\frac{1980 + 990 + 660 + 495}{1980} - \frac{220 + 198 + 180 + 165}{1980} \right) \\
 &= 5 \left(\frac{4125}{1980} - \frac{763}{1980} \right) = \frac{3362}{396} = \boxed{\frac{1681}{198}}.
 \end{aligned}$$

- T-14. Let N be the six-digit number with the property described in the problem. Let a_0 be the rightmost digit of N such that $N = 10k + a_0$, where $10000 \leq k \leq 99999$. Therefore $4N = 10^5 a_0 + k$, and combining the two equations gives $4(10k + a_0) = 10^5 a_0 + k \rightarrow 39k = 99996a_0 \rightarrow k = 2564a_0$. k is greater than or equal to 10000 for the first time when $a_0 = 4$, and the resulting six-digit number is $\boxed{102564}$.

T-15. By the Chinese Remainder Theorem, it suffices to look modulo 2 and modulo 7. Modulo 2, there are clearly only two solutions because x_k must be even for $k = 1, 2, \dots, 13$.

Modulo 7, there are two possible situations:

- The values of $x_k^2 \pmod{7}$ cycle between 1, 4, 2 in that order. There are 3 ways to choose the value x_1 modulo 7, and afterwards for each k there are 2 ways to choose x_k based on x_{k-1} . Hence $3 \cdot 2^{14}$ sequences in this case.
- The value $x_k^2 \pmod{7}$ is always zero. There is only 1 sequence in this case, the constant zero sequence.

So the final answer is $2(3 \cdot 2^{14} + 1) = \boxed{98306}$.